Ryota Takatsuki

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Google Scholar | GitHub | LinkedIn

Education

Sussex Centre for Consciousness Science, University of Sussex (Brighton, UK) —*Ph.D. in Informatics* Sep 2025 –Mar 2029 (planned)

- Supervised by Prof. Anil Seth and Prof. Christopher Buckley
- Thesis: interpretability-driven consciousness science —establishing consciousness-associated perceptual phenomena (bistable perception, postdictive integration, illusory contours) behaviorally in vision, language, and multimodal models, then using mechanistic interpretability to characterize how they are implemented

The University of Tokyo (Tokyo, Japan) —*M.A. program in Applied Computer Science, Interdisciplinary Information Studies* Apr 2023 –Sep 2025

- Developed a lightweight real-time 3D video transmission system for drone holoportation applications, supervised by Prof. Akihiro Nakao
- Withdrew in Sep 2025 to begin Ph.D. studies after a gap year since Apr 2024

The University of Tokyo (Tokyo, Japan) —*B.S. in Systems Innovation, Engineering* Apr 2019 –Mar 2023

- Thesis: High-presence communication through Local 5G ultra-low latency transmission of RGBD images (supervised by Prof. Akihiro Nakao)

Experience

AI Alignment Network (Tokyo, Japan) —*Research Fellow*, Jun 2024 –present

- Community building for AI alignment: hosting a mechanistic interpretability study group, writing educational blogs

R&D Center for Large Language Models, National Institute of Informatics (Tokyo, Japan) —*Technical Assistant*, Jun 2024 –Sep 2025

- Bistable perception in vision language models, in collaboration with Yanaka Laboratory (UTokyo)

Araya Inc. (Tokyo, Japan) —*Moonshot program R&D intern*, Dec 2021 –Sep 2025

- Supervised by Dr. Ippei Fujisawa and Dr. Ryota Kanai
- Mechanistic interpretability of optical illusions in transformer-based models and interpretability tooling, in collaboration with Sonia Joseph (McGill

Univ.); meta-networks in the context of AI interpretability and consciousness

The University of Tokyo —*Tutor for international students*, Apr 2023 –Mar 2024

Selected publications

- **Takatsuki, R.**, Doi, T., Watahiki, A., Seth, A. K., & Yanaka, H. (2026). (How) Do MLLMs Report Bistable Images Like Humans? *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*, main conference, Budapest, Hungary, Oct 24–29, 2026. (To appear)
- Aoki, K., **Takatsuki, R.**, Minegishi, G., Haruki, Y., & Kawahara, D. (2026). In-Context Neurofeedback: Can LLMs Control Their Internal Representations through Privileged Access? *Findings of the Association for Computational Linguistics: EMNLP 2026*, Budapest, Hungary, Oct 24–29, 2026. (To appear.) Earlier version: *Trustworthy AI for Good (AI4GOOD) Workshop @ ICML 2026*, Seoul, South Korea. (Spotlight)
- Aoki, K., **Takatsuki, R.**, Minegishi, G., & Cho, H. (2026). Mechanistic Interpretability: A New Trend in Interpretability Research (in Japanese). *Transactions of the Japanese Society for Artificial Intelligence*, 41(4), 1–13 (40th-anniversary special issue). JSAI 40th Anniversary Commemorative Paper Award —Best Paper.
- Kanai, R., **Takatsuki, R.**† (†corresponding author), & Fujisawa, I. (2025). Meta-representations as representations of processes. *Neuroscience of Consciousness*, 2025(1), niaf038.
- **Takatsuki, R.**, Joseph, S., Fujisawa, I., & Kanai, R. (2025). Decoding Vision Transformers: the Diffusion Steering Lens. *The First Workshop on Mechanistic Interpretability for Vision, CVPR 2025* (IEEE CVPRW 2025 proceedings, pp. 4819–4824).

Full publication list

Journal articles

- Aoki, K., **Takatsuki, R.**, Minegishi, G., & Cho, H. (2026). Mechanistic Interpretability: A New Trend in Interpretability Research (in Japanese). *Transactions of the Japanese Society for Artificial Intelligence*, 41(4), 1–13 (40th-anniversary special issue). JSAI 40th Anniversary Commemorative Paper Award —Best Paper.
- Kanai, R., **Takatsuki, R.**† (†corresponding author), & Fujisawa, I. (2025). Meta-representations as representations of processes. *Neuroscience of Consciousness*, 2025(1), niaf038.

Conference and workshop papers

- **Takatsuki, R.**, Doi, T., Watahiki, A., Seth, A. K., & Yanaka, H. (2026). (How) Do MLLMs Report Bistable Images Like Humans? *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*, main conference, Budapest, Hungary, Oct 24–29, 2026. (To appear)
- Aoki, K., **Takatsuki, R.**, Minegishi, G., Haruki, Y., & Kawahara, D. (2026). In-Context Neurofeedback: Can LLMs Control Their Internal Representations through Privileged Access? *Findings of the Association for Computational Linguistics: EMNLP 2026*, Budapest, Hungary, Oct 24–29, 2026. (To appear.) Earlier version: *Trustworthy AI for Good (AI4GOOD) Workshop @ ICML 2026*, Seoul, South Korea. (Spotlight)
- **Takatsuki, R.**, Joseph, S., Fujisawa, I., & Kanai, R. (2025). Decoding Vision Transformers: the Diffusion Steering Lens. *The First Workshop on Mechanistic Interpretability for Vision, CVPR 2025* (IEEE CVPRW 2025 proceedings, pp. 4819–4824).

Technical reports

- **Takatsuki, R.**, Kanai, K., & Nakao, A. (2024). A real-time 3D video transmission and display system for drone holoportation applications. *IEICE Tech. Rep.*, 123(367), 13–18. (Encouragement Talk)
- **Takatsuki, R.**, & Nakao, A. (2023). High-presence communication through Local 5G ultra-low latency transmission of RGBD images. *IEICE Tech. Rep.*, 122(406), 157–162.
- **Takatsuki, R.**, & Nakao, A. (2023). Quantitative evaluation of the impact of the inter-access point interference on unlicensed communication in high-capacity, low-latency transmission. *IEICE Tech. Rep.*, 122(406), 227–232.

Other articles

- **Takatsuki, R.**, & Yamakawa, H. (2025). Conference report: The 39th Annual AAAI Conference on Artificial Intelligence (AAAI-25). 人工知能 *(Journal of JSAI)*, 40(4), 615–616.
- Alon, N., Annunziata, D., Bouneffouf, D., Burstein, J., Canzaniello, M., Chaudhri, V. K., Deshwal, A., Esmaeili, S. A., Lin, B., Liu, Z., Lu, Y.-H., Basu Mallick, D., Olney, A. M., **Takatsuki, R.**, Wang, P., Woodhead, S., & Yang, Q. (2025). Reports of the Workshops Held at the 2025 AAAI Conference on Artificial Intelligence. *Interactive AI Magazine*, June 9, 2025. (Authors alphabetical; I wrote the report on workshop W18, Post-Singularity Symbiosis.)

Presentations

Oral

- **Takatsuki, R.**, Joseph, S., Fujisawa, I., & Kanai, R. "Towards Reverse-engineering the Kanizsa Illusion: A Mechanistic Study of Vision Transformer". *JSAI 2025*, Osaka, Japan. May 29, 2025.
- Aoki, K., **Takatsuki, R.**, & Minegishi, G. "Mechanistic Interpretability: A New Trend in Interpretability Research". *JSAI 2025*, Osaka, Japan. May 29, 2025.
- **Takatsuki, R.**, Fujisawa, I., & Kanai, R. "Towards elucidating the mechanism of illusory contours through mechanistic interpretability". *Vision Society of Japan*, Tokyo, Japan. Sep 15, 2024.
- **Takatsuki, R.**, Fujisawa, I., & Kanai, R. "Qualia as Meta-Representations of Processes". *ASSC 27*, Tokyo, Japan. Jul 4, 2024.

Poster

- **Takatsuki, R.**, Nishimoto, S., Minegishi, G., & Seth, A. K. "Do Autoregressive LLMs Exhibit Postdiction-Like Integration of Later Tokens?" . *ASSC 29*, Santiago, Chile. Jul 1, 2026. (Poster P1-021)
- **Takatsuki, R.**, Fujisawa, I., & Kanai, R. "Meta-Representations and Conceptual IIT". *Post-ASSC satellite symposium: Mathematical and philosophical models of consciousness*, Kyoto, Japan. Jul 8, 2024.
- **Takatsuki, R.**, Fujisawa, I., & Kanai, R. "Weight structures in ANNs and their modality-specificity". *Consciousness Research Network 2023*, Taipei, Taiwan. Aug 26, 2023.

Invited

- **Takatsuki, R.** "Introduction to Mechanistic Interpretability". *Student Session, International Conference on Robot Ethics and Standards 2024*. Jul 31, 2024.

Service

- Reviewer, The First Workshop on Mechanistic Interpretability for Vision, CVPR 2025
- Organizing Committee, Organized Session "Mechanistic Interpretability", JSAI 2025
- Organizing Committee and Reviewer, 1st Workshop on Post-Singularity Symbiosis: Preparing for a World with Superintelligence (PSS 2025), AAAI-25

Awards

- **JSAI 40th Anniversary Commemorative Paper Award —Best Paper (最優秀論文)**, Category A "Survey papers pioneering the future of AI",

Japanese Society for Artificial Intelligence, 2026 —for "Mechanistic Interpretability: A New Trend in Interpretability Research"
- **ASSC 29 Travel Award**, Association for the Scientific Study of Consciousness, 2026
- **Sussex Centre for Consciousness Science PhD Studentship**, University of Sussex, 2025–2029 —stipend and fee waiver for 3.5 years, plus a research grant
- **Academist Prize 5th** —"Understanding the World as Perceived by AI", 2025 —crowdfunding research grant of ~2 million JPY from 238 supporters, plus an additional 300k JPY grant
- **Young Researcher Encouragement Award**, IEICE Technical Committee on Network Systems, 2024 —given to ~10% of eligible young-researcher presentations

Other activities

- 2025 QStr-IIT Summer School (ATR, Tiny Blue Dot Foundation, and Neuroscience School of Advanced Studies), San Servolo (Venice) and Bertinoro, Italy, Oct 5–17, 2025 (partially funded) —two-week residential program on the qualia-structure paradigm, integrated information theory, and the neural basis of consciousness
- Autumn School of Computational Neuroscience (ASCONE), Nov 2024 (fully funded)
- Neuromatch Academy, Computational Neuroscience, Jul 2024
- Host, weekly Mechanistic Interpretability Study Group, AI Alignment Network, May 2024 –Apr 2025
- ITU delegate, Huawei Asia-Pacific Seeds for the Future Program 2022, Thailand (fully funded)
- Japanese delegate, ITU Generation Connect Global Youth Summit 2022, Kigali, Rwanda (fully funded)

Language proficiency

- Japanese: native
- English: IELTS Academic Overall 8.0 (Dec 2021)
- Mandarin Chinese: conversational